

TE Curve Verification Pipeline Familywise ONNX Report - wave4_4_causal_tcn_latent_offset_residual

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml1`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave4_4_causal_tcn_latent_offset_residual/2026-07-15-23-45-15__track2_wave4_4_causal_tcn_latent_offset_residual_familywise_onnx_report` ;
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave4_4_causal_tcn_latent_offset_residual/2026-07-15-23-45-15__track2_wave4_4_causal_tcn_latent_offset_residual_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_4_causal_tcn_latent_offset_residual/2026-07-15-23-45-15__track2_wave4_4_causal_tcn_latent_offset_residual_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_4_causal_tcn_latent_offset_residual/2026-07-15-23-45-15__track2_wave4_4_causal_tcn_latent_offset_residual_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave4_4_causal_tcn_latent_offset_residual` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave4_4_causal_tcn_latent_offset_residual_fw_simplified_setpoints</code>	<code>2026-07-15-20-14-26_te_wave4_4_causal_tcn_latent_offset_residual_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave4_4_causal_tcn_latent_offset_residual_bw_simplified_setpoints</code>	<code>2026-07-15-20-38-50_te_wave4_4_causal_tcn_latent_offset_residual_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave4_4_causal_tcn_latent_offset_residual_global_simplified_setpoints</code>	<code>2026-07-15-20-07-16_te_wave4_4_causal_tcn_latent_offset_residual_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/forward/2026-07-15-20-14-26_te_wave4_4_causal_tcn_latent_offset_residual_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/forward/2026-07-15-20-14-26_te_wave4_4_causal_tcn_latent_offset_residual_fw_simplified_setpoints/python/latent_state_hysteresis_probe-epoch=236-val_mae=0.00349844.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/backward/2026-07-15-20-38-50_te_wave4_4_causal_tcn_latent_offset_residual_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/backward/2026-07-15-20-38-50_te_wave4_4_causal_tcn_latent_offset_residual_bw_simplified_setpoints/python/latent_state_hysteresis_probe-epoch=105-val_mae=0.00370934.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/global/2026-07-15-20-07-16_te_wave4_4_causal_tcn_latent_offset_residual_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/global/2026-07-15-20-07-16_te_wave4_4_causal_tcn_latent_offset_residual_global_simplified_setpoints/python/latent_state_hysteresis_probe-epoch=055-val_mae=0.00376625.ckpt</code>

Aggregate Metrics

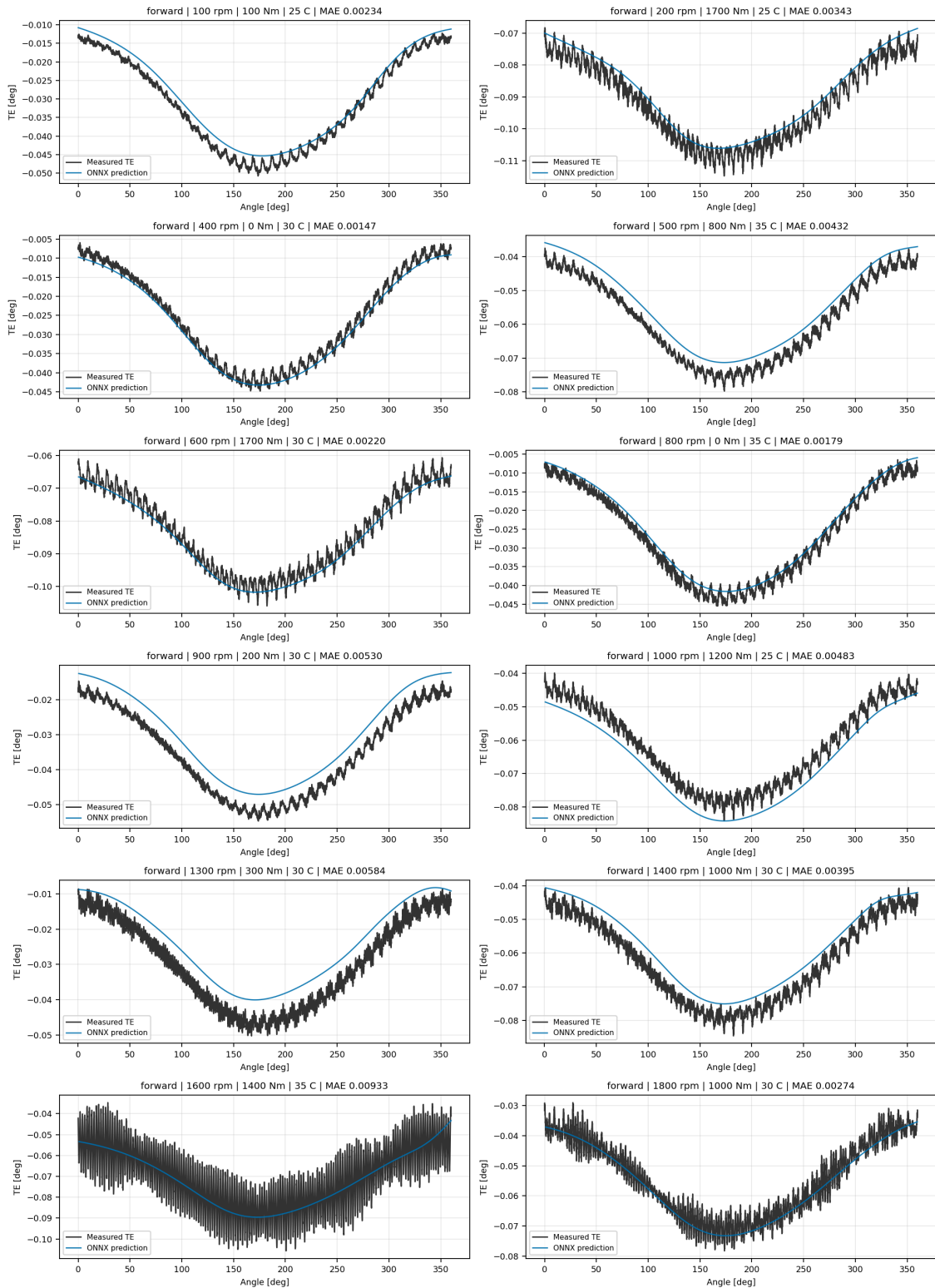
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003261	0.003690	7.650	14.507
backward	97	0.003713	0.004153	8.641	16.030
global	194	0.003548	0.003989	8.280	14.373

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000012	0.002621	0.001748	0.006147
backward	0.000398	0.002988	0.001910	0.004800
global	-0.000035	0.002891	0.001838	0.006096

Forward 12-Curve Page

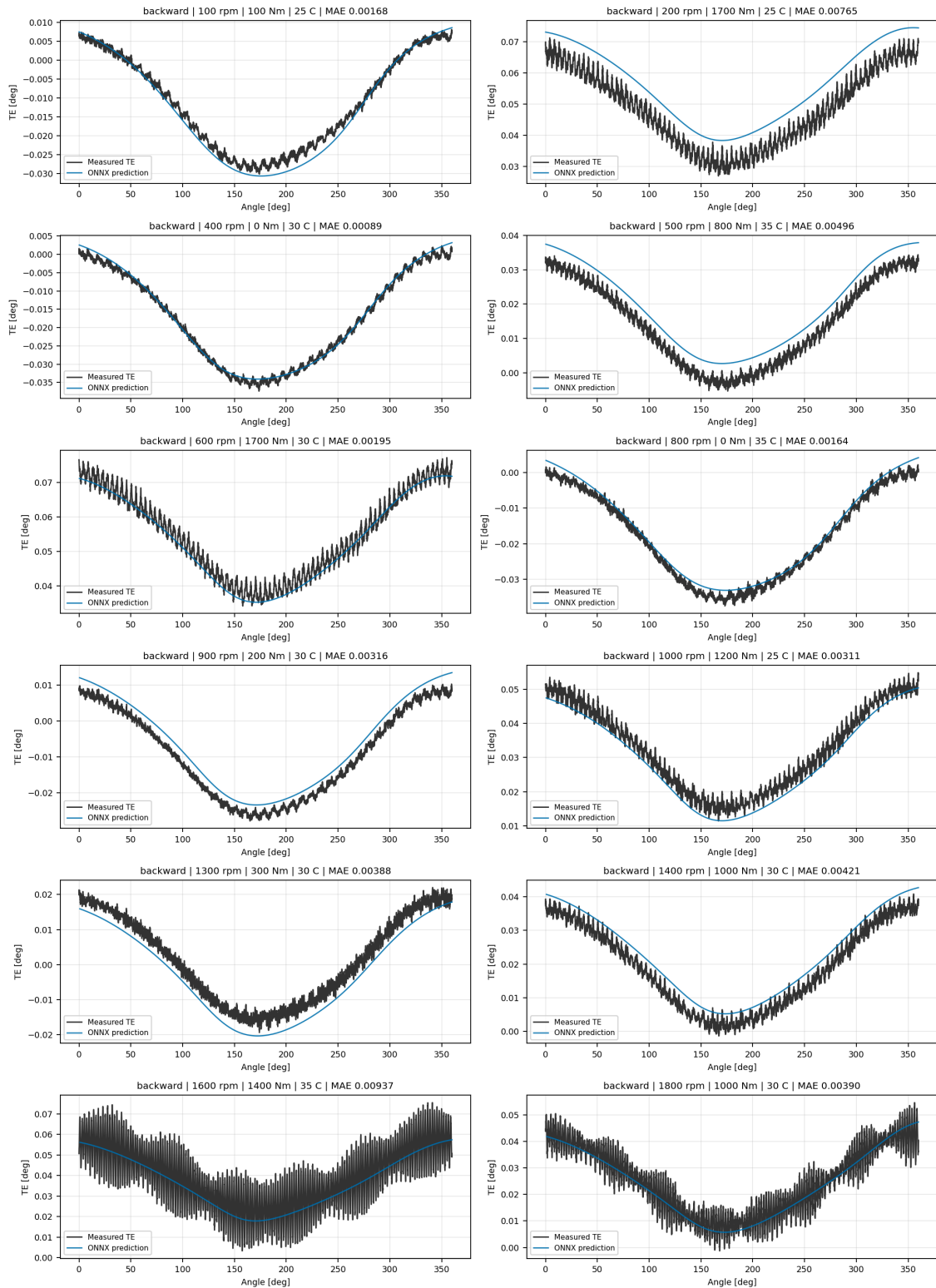
Simplified Dataset + Setpoints | wave4_4_causal_tcn_latent_offset_residual | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

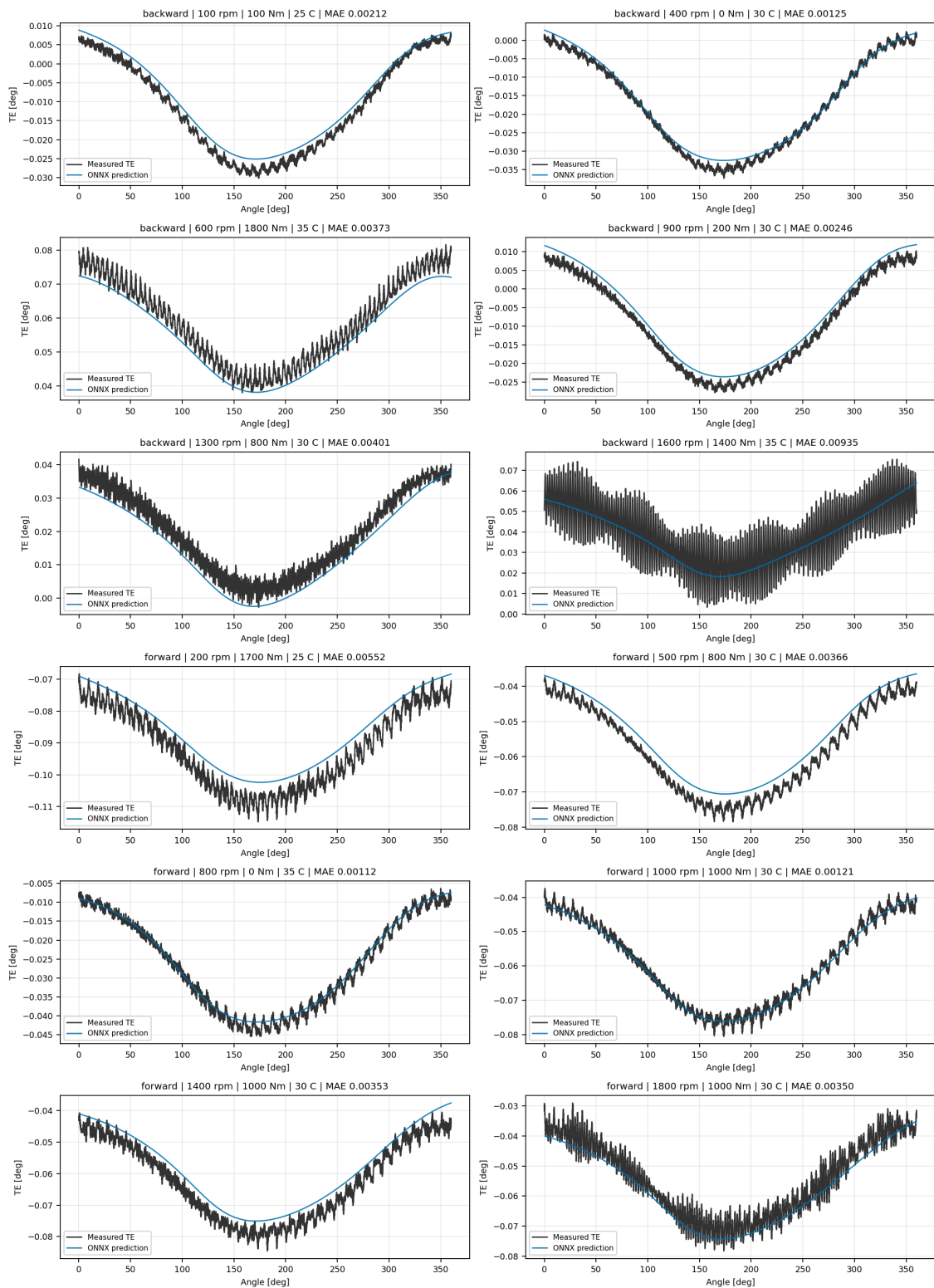
Simplified Dataset + Setpoints | wave4_4_causal_tcn_latent_offset_residual | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave4_4_causal_tcn_latent_offset_residual | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave4_4_causal_tcn_latent_offset_residual ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_4_causal_tcn_latent_offset_residual_fw_polished_setpoints	2026-07-15-21-27-08_te_wave4_4_causal_tcn_latent_offset_residual_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave4_4_causal_tcn_latent_offset_residual_bw_polished_setpoints	2026-07-15-21-50-42_te_wave4_4_causal_tcn_latent_offset_residual_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave4_4_causal_tcn_latent_offset_residual_global_polished_setpoints	2026-07-15-21-10-35_te_wave4_4_causal_tcn_latent_offset_residual_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/forward/2026-07-15-21-27-08_te_wave4_4_causal_tcn_latent_offset_residual_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/forward/2026-07-15-21-27-08_te_wave4_4_causal_tcn_latent_offset_residual_fw_polished_setpoints/python/latent_state_hysteresis_probe-epoch=107-val_mae=0.00221473.ckpt
backward	models/polished_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/backward/2026-07-15-21-50-42_te_wave4_4_causal_tcn_latent_offset_residual_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/backward/2026-07-15-21-50-42_te_wave4_4_causal_tcn_latent_offset_residual_bw_polished_setpoints/python/latent_state_hysteresis_probe-epoch=074-val_mae=0.00224017.ckpt
global	models/polished_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/global/2026-07-15-21-10-35_te_wave4_4_causal_tcn_latent_offset_residual_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_4_causal_tcn_latent_offset_residual/global/2026-07-15-21-10-35_te_wave4_4_causal_tcn_latent_offset_residual_global_polished_setpoints/python/latent_state_hysteresis_probe-epoch=060-val_mae=0.00222789.ckpt

Aggregate Metrics

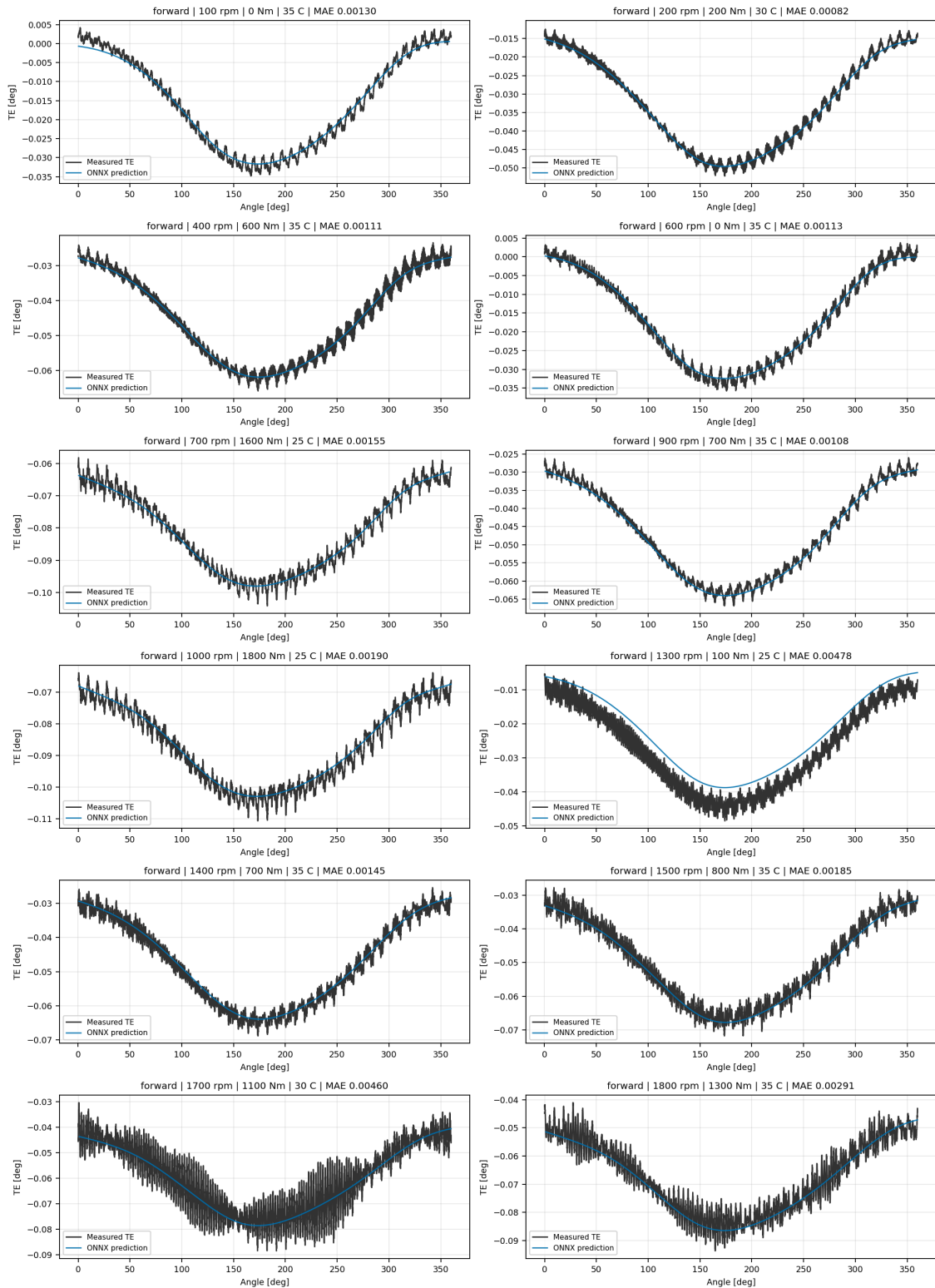
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002307	0.002763	5.148	11.986
backward	94	0.002784	0.003302	5.388	11.522
global	194	0.002515	0.003019	5.204	11.393

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000047	0.001003	0.001918	0.006548
backward	0.000476	0.000961	0.002414	0.006619
global	0.000001	0.001042	0.002165	0.006338

Forward 12-Curve Page

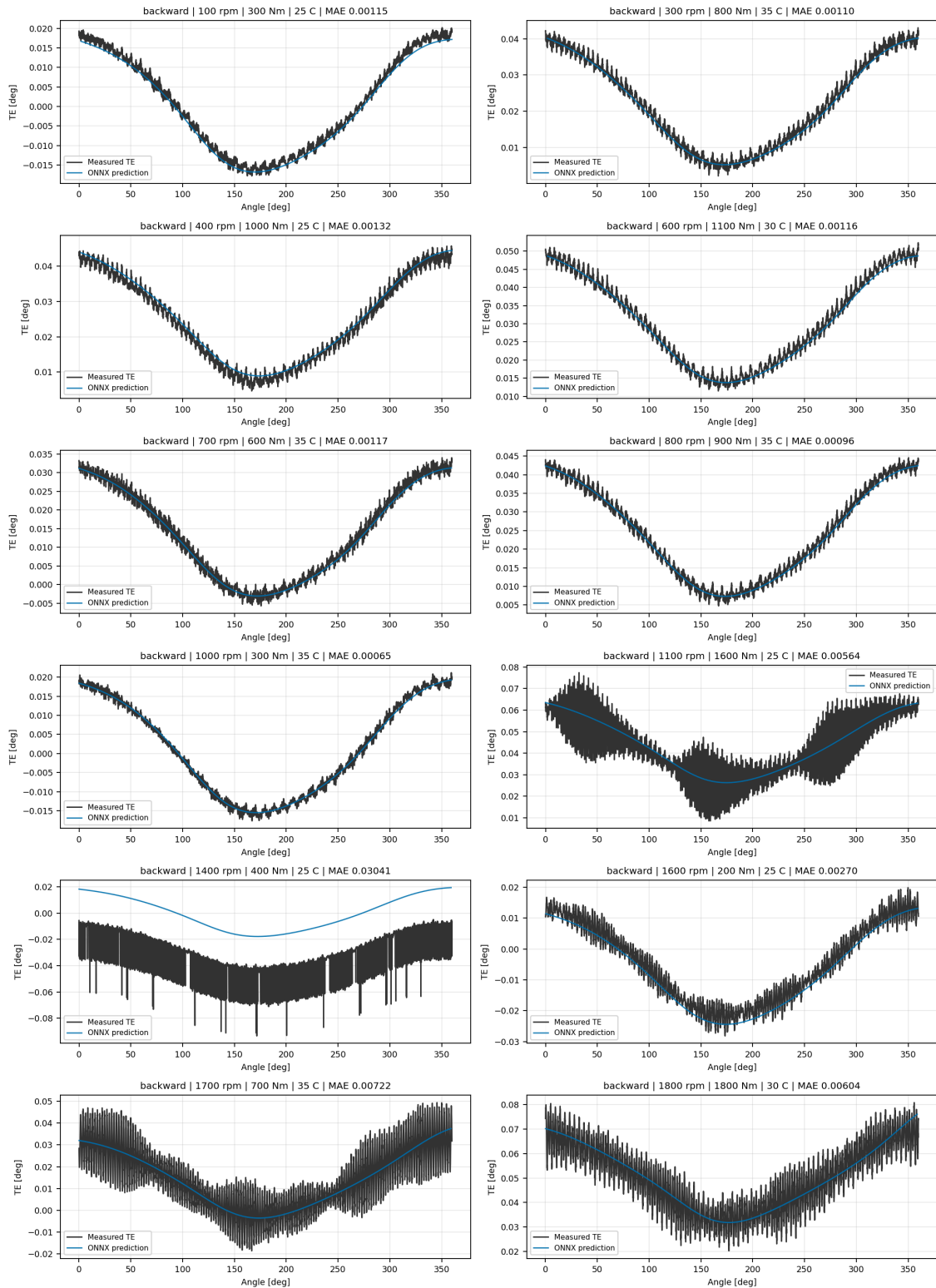
Polished Dataset + Setpoints | wave4_4_causal_tcn_latent_offset_residual | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

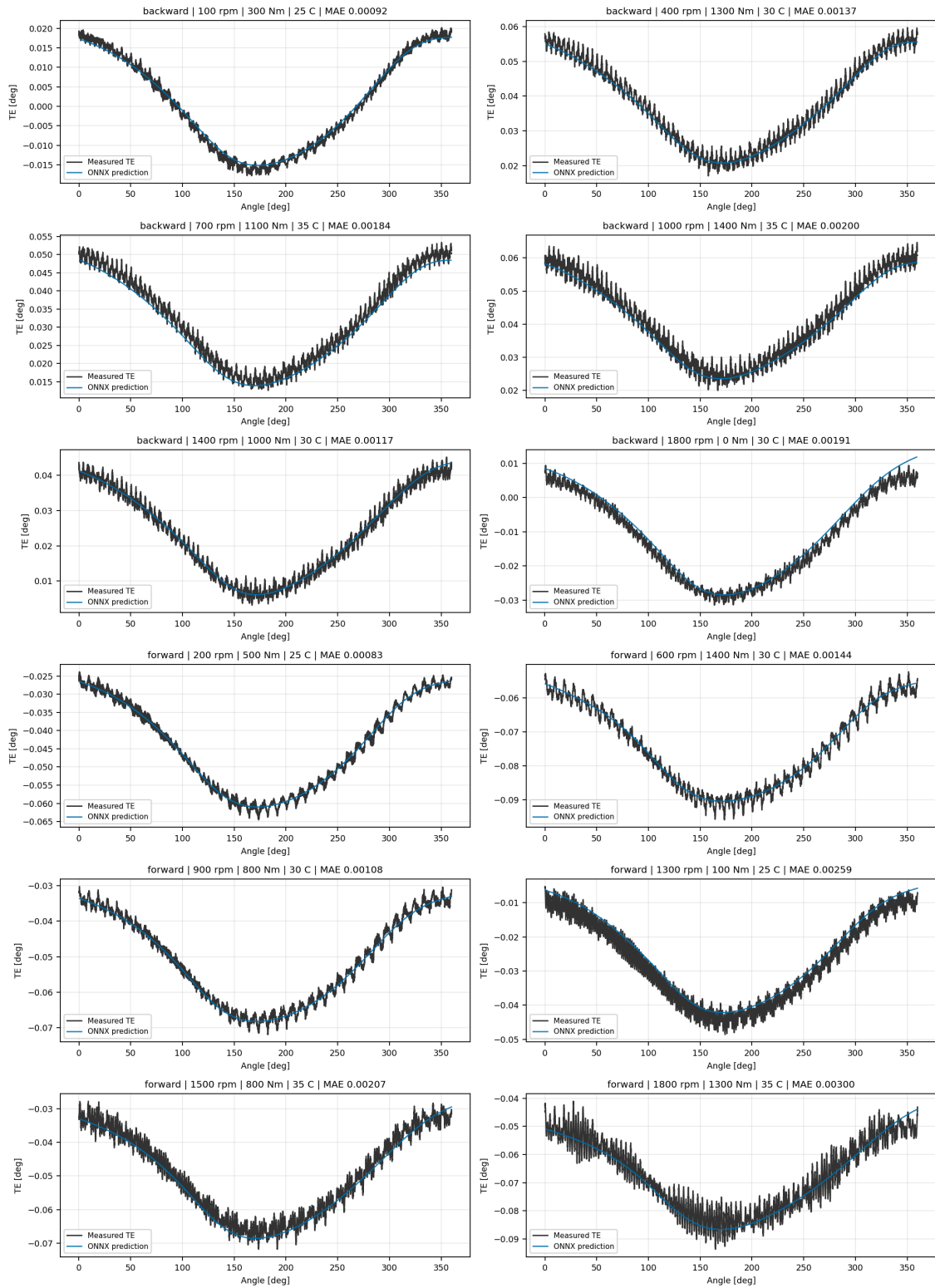
Polished Dataset + Setpoints | wave4_causal_tcn_latent_offset_residual | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave4_causal_tcn_latent_offset_residual | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave4_4_causal_tcn_latent_offset_residual ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_4_causal_tcn_latent_offset_residual_fw_polished_actual_values	2026-07-15-22-40-16_te_wave4_4_causal_tcn_latent_offset_residual_fw_polished_actual_values	polished_point_v1
backward	te_wave4_4_causal_tcn_latent_offset_residual_bw_polished_actual_values	2026-07-15-22-54-23_te_wave4_4_causal_tcn_latent_offset_residual_bw_polished_actual_values	polished_point_v1
global	te_wave4_4_causal_tcn_latent_offset_residual_global_polished_actual_values	2026-07-15-22-27-43_te_wave4_4_causal_tcn_latent_offset_residual_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave4_4_causal_tcn_latent_offset_residual/forward/2026-07-15-22-40-16_te_wave4_4_causal_tcn_latent_offset_residual_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_4_causal_tcn_latent_offset_residual/forward/2026-07-15-22-40-16_te_wave4_4_causal_tcn_latent_offset_residual_fw_polished_actual_values/python/latent_state_hysteresis_probe-epoch=038-val_mae=0.00225416.ckpt
backward	models/polished_dataset/actual_values/exported/wave4_4_causal_tcn_latent_offset_residual/backward/2026-07-15-22-54-23_te_wave4_4_causal_tcn_latent_offset_residual_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_4_causal_tcn_latent_offset_residual/backward/2026-07-15-22-54-23_te_wave4_4_causal_tcn_latent_offset_residual_bw_polished_actual_values/python/latent_state_hysteresis_probe-epoch=078-val_mae=0.00222665.ckpt
global	models/polished_dataset/actual_values/exported/wave4_4_causal_tcn_latent_offset_residual/global/2026-07-15-22-27-43_te_wave4_4_causal_tcn_latent_offset_residual_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_4_causal_tcn_latent_offset_residual/global/2026-07-15-22-27-43_te_wave4_4_causal_tcn_latent_offset_residual_global_polished_actual_values/python/latent_state_hysteresis_probe-epoch=041-val_mae=0.00230429.ckpt

Aggregate Metrics

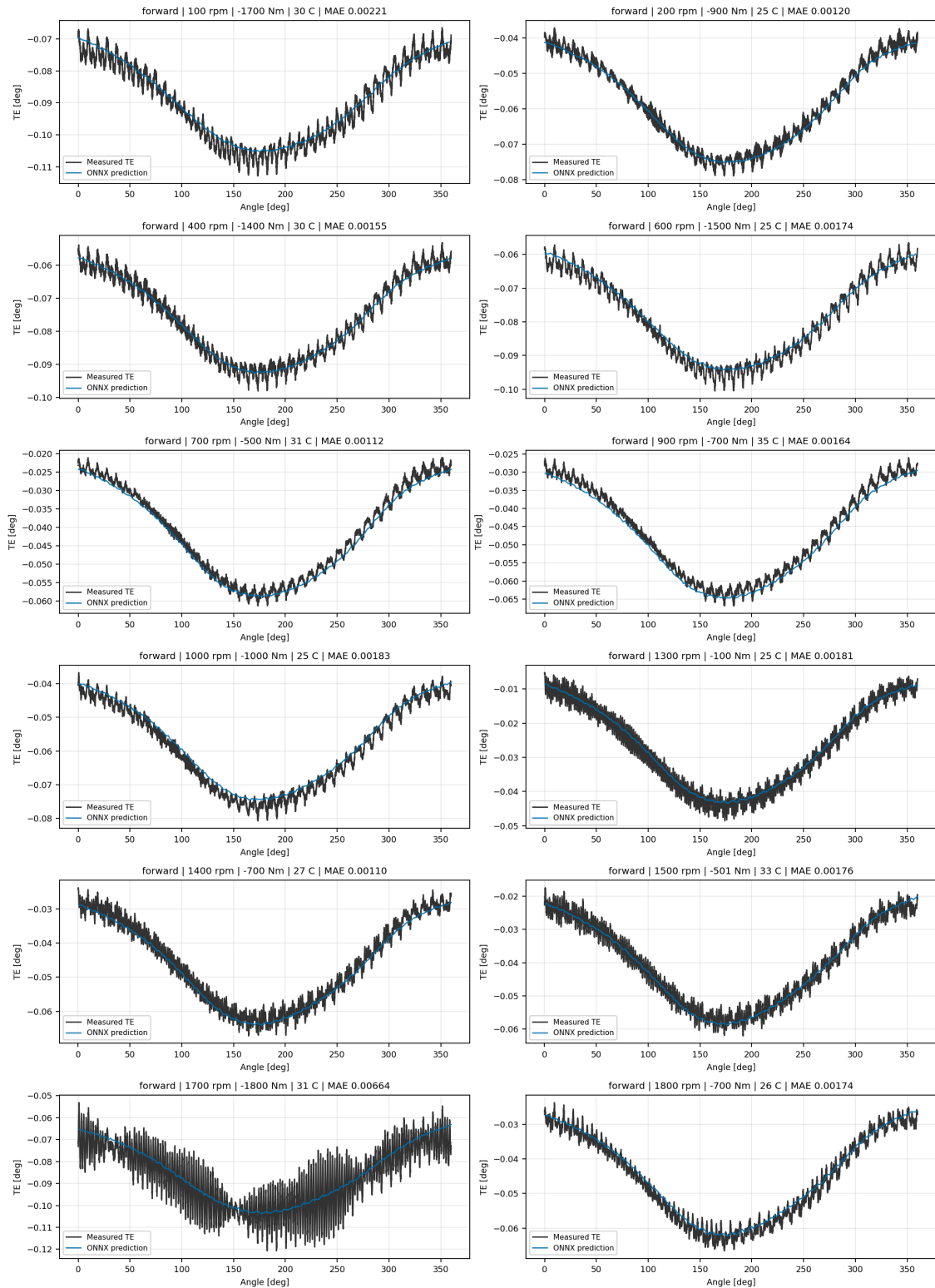
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002209	0.002680	4.882	10.614
backward	94	0.002529	0.003079	4.973	11.654
global	194	0.002420	0.002956	5.053	11.649

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000333	0.000863	0.001972	0.006472
backward	-0.000241	0.000602	0.002437	0.005110
global	-0.000282	0.000829	0.002230	0.006169

Forward 12-Curve Page

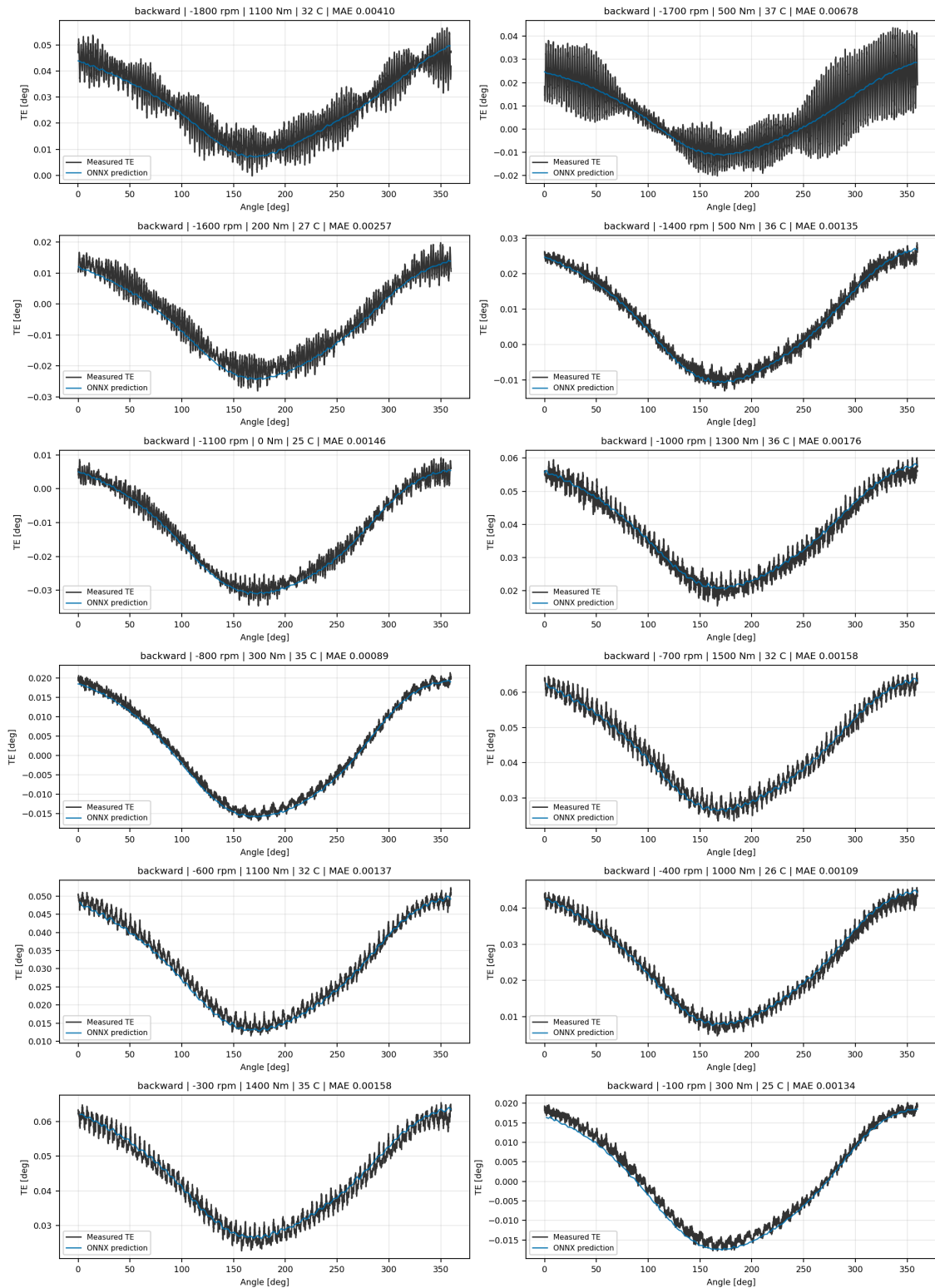
Polished Dataset + Actual Values | wave4_4_causal_tcn_latent_offset_residual | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

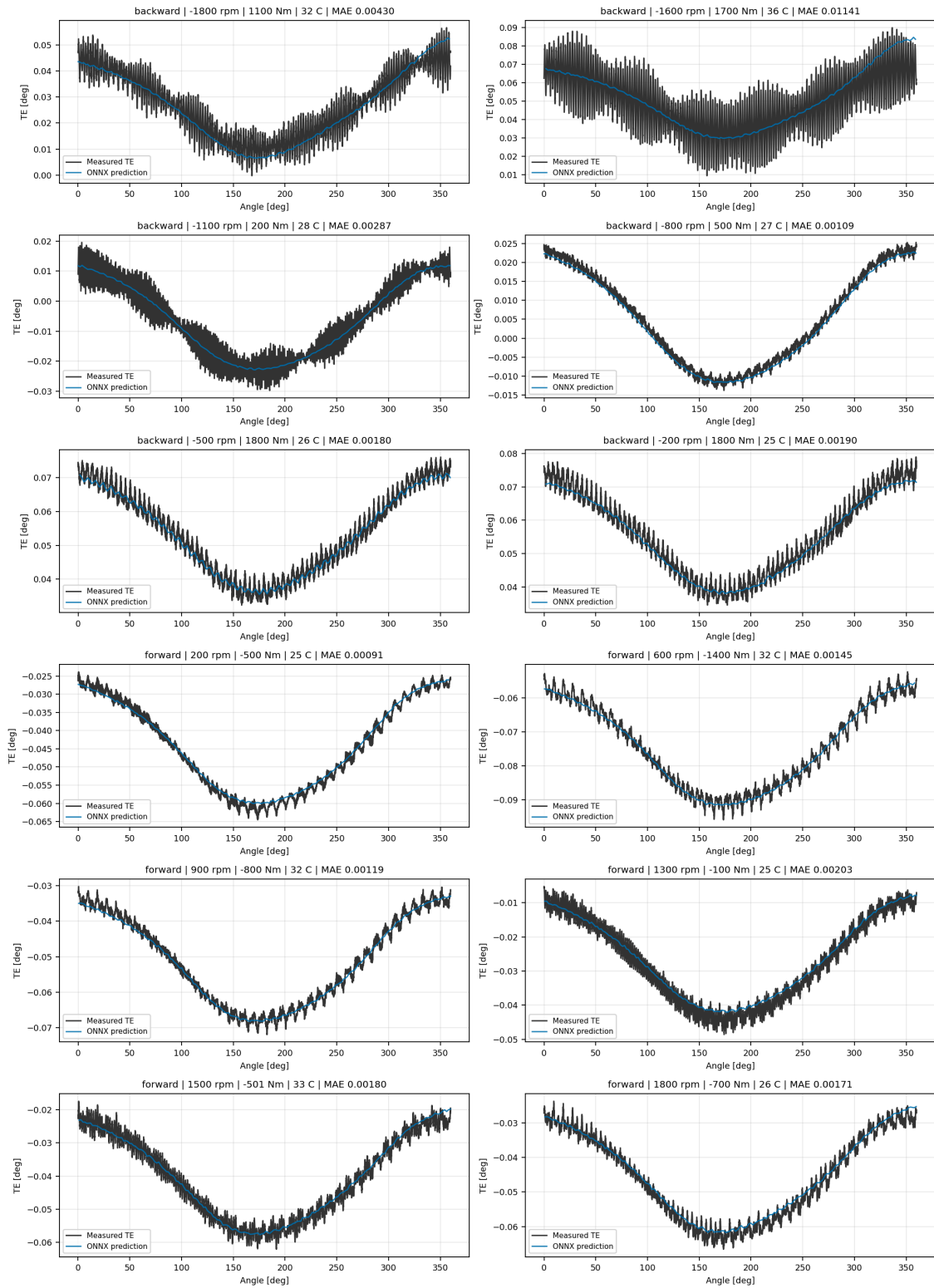
Polished Dataset + Actual Values | wave4_causal_tcn_latent_offset_residual | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave4_causal_tcn_latent_offset_residual | global



polished_dataset_actual_values global 12-curve collage